Block practice · More nerves, senses, hormones and drugs

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

State causes and symptoms of dementia, Parkinson's and mental illness

Identify ear parts and use the sound path to explain hearing and hearing loss

Show and explain why impulses travel one way across a synapse

OLevels.com SCHOOL

Practice workbook · State causes and symptoms of dementia, Parkinson's and mental illness

Recalling or extracting from a passage or graph the symptoms, causes, risk factors and evidence for Alzheimer's, Parkinson's, dementia, depression and schizophrenia.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-RECALL-BRAIN-DISORDER-FEATURES

1. Question 6 · part c · 2 marks

June 2019 Paper 2R

Worked-example retry · Attempt part c only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Alzheimer's disease is a degenerative disease that usually starts slowly and gets worse over time. Scientists want to understand the causes of the disease and discover treatments for the memory loss that is a major feature of Alzheimer's. Scientists use the brains of people who have died to try to find out what happens in the brains of people who have Alzheimer's.

The most common early symptom is short-term memory loss, which is difficulty in remembering recent events. People with this condition are no longer able to transfer information about their daily life from their short-term to long-term memory. For example, they are unable to remember if they went shopping.

The condition causes plaques of an insoluble protein called B-amyloid to build up in the gap between neurones. Over time, these plaques appear in the neurones themselves in the regions of the brain associated with memory.

The B-amyloid is formed when an essential protein called amyloid precursor protein (APP) is not broken down correctly.

(a) Describe two differences between short-term memory and long-term memory (lines 8 to 9). **(2)**

1 .....

.....

2 .....

.....

(b) State the area of the brain responsible for memory. **(1)**

.....

(c) State two symptoms of Alzheimer's disease, other than memory loss. **(2)**

1 .....

.....

2 .....

.....

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OLevels.com SCHOOL

16

P 6 0 1 8 7 A 0 1 6 2 4

1. Question 6 · part c · 2 marks

June 2019 Paper 2R

Worked-example retry · Attempt part c only.

(d) Explain why the build-up of B-amyloid disrupts brain function (lines 10 to 11).

(3)

(e) One method of treating the early stages of Alzheimer's is the use of an enzyme inhibitor.

Explain how this treatment could help people with this condition.

(2)

(f) Describe the evidence that shows the amyloid precursor protein (APP) is involved in the development of Alzheimer's disease (lines 10 to 14).

(2)

(Total for Question 6 = 12 marks)

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1. Question 6 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 6 · part a · 2 marks

June 2019 Paper 2R

Further practice · Attempt part a only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Alzheimer's disease is a degenerative disease that usually starts slowly and gets worse over time. Scientists want to understand the causes of the disease and discover treatments for the memory loss that is a major feature of Alzheimer's. Scientists use the brains of people who have died to try to find out what happens in the brains of people who have Alzheimer's.

The most common early symptom is short-term memory loss, which is difficulty in remembering recent events. People with this condition are no longer able to transfer information about their daily life from their short-term to long-term memory. For example, they are unable to remember if they went shopping.

The condition causes plaques of an insoluble protein called B-amyloid to build up in the gap between neurones. Over time, these plaques appear in the neurones themselves in the regions of the brain associated with memory.

The B-amyloid is formed when an essential protein called amyloid precursor protein (APP) is not broken down correctly.

(a) Describe two differences between short-term memory and long-term memory (lines 8 to 9). **(2)**

1 .....

.....

2 .....

.....

(b) State the area of the brain responsible for memory. **(1)**

.....

(c) State two symptoms of Alzheimer's disease, other than memory loss. **(2)**

1 .....

.....

2 .....

.....

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OLevels.com SCHOOL

16

P 6 0 1 8 7 A 0 1 6 2 4

2. Question 6 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 6 · part f · 2 marks

June 2019 Paper 2R

Further practice · Attempt part f only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Alzheimer's disease is a degenerative disease that usually starts slowly and gets worse over time. Scientists want to understand the causes of the disease and discover treatments for the memory loss that is a major feature of Alzheimer's. Scientists use the brains of people who have died to try to find out what happens in the brains of people who have Alzheimer's.

The most common early symptom is short-term memory loss, which is difficulty in remembering recent events. People with this condition are no longer able to transfer information about their daily life from their short-term to long-term memory. For example, they are unable to remember if they went shopping.

The condition causes plaques of an insoluble protein called B-amyloid to build up in the gap between neurones. Over time, these plaques appear in the neurones themselves in the regions of the brain associated with memory.

The B-amyloid is formed when an essential protein called amyloid precursor protein (APP) is not broken down correctly.

(a) Describe two differences between short-term memory and long-term memory (lines 8 to 9). **(2)**

1 .....

.....

2 .....

.....

(b) State the area of the brain responsible for memory. **(1)**

.....

(c) State two symptoms of Alzheimer's disease, other than memory loss. **(2)**

1 .....

.....

2 .....

.....

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OLevels.com SCHOOL

16

3. Question 6 · part f · 2 marks

June 2019 Paper 2R

Further practice · Attempt part f only.

(d) Explain why the build-up of B-amyloid disrupts brain function (lines 10 to 11).

(3)

(e) One method of treating the early stages of Alzheimer's is the use of an enzyme inhibitor.

Explain how this treatment could help people with this condition.

(2)

(f) Describe the evidence that shows the amyloid precursor protein (APP) is involved in the development of Alzheimer's disease (lines 10 to 14).

(2)

(Total for Question 6 = 12 marks)

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3. Question 6 · part f · 2 marks · Working and answer

Working and answer

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4. Question 5 · part b.i · 2 marks

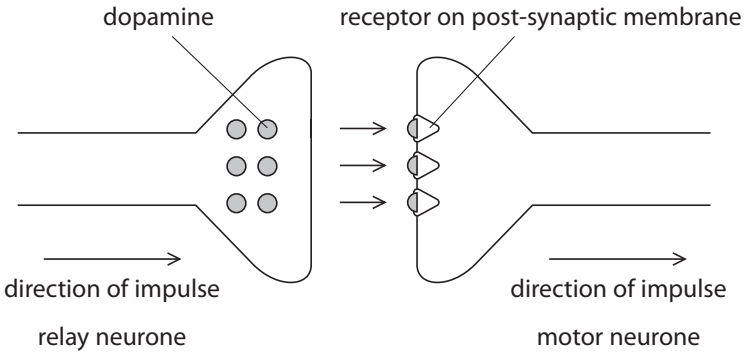
June 2023 Paper 2R

Further practice · Attempt part b.i only.

5 (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.



(i) Describe what is meant by a neurotransmitter.

(2)

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

(2)

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P 7 2 5 8 8 A 0 1 4 2 4

4. Question 5 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

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(iii) Suggest how an excess of dopamine will affect bodily functions.

(2)

.....

.....

.....

.....

(b) In 2017 in the United Kingdom, 145 000 people were diagnosed with Parkinson's disease.

This figure is expected to have an 18% increase by 2025.

(i) Give two symptoms of Parkinson's disease.

(2)

1 .....

.....

2 .....

.....

(ii) Calculate the number of people expected to be diagnosed with Parkinson's disease in 2025.

(2)

number of people = ...

(Total for Question 5 = 10 marks)

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4. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part c.iv · 2 marks

November 2024 Paper 2

Further practice · Attempt part c.iv only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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5. Question 4 · part c.iv · 2 marks

November 2024 Paper 2

Further practice · Attempt part c.iv only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

5. Question 4 · part c.iv · 2 marks

November 2024 Paper 2

Further practice · Attempt part c.iv only.

(iii) Calculate the ratio of the number of people with non-infectious diseases in a higher income country to the number of people with non-infectious diseases in a lower income country (lines 20 and 21).

Give your answer in the form n : 1

(3)

ratio = ... : .....

(iv) Suggest why there are more cases of dementia and cancers in higher income countries (lines 21 and 22).

(2)

.....

.....

.....

.....

.....

.....

.....

(Total for Question 4 = 15 marks)

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5. Question 4 · part c.iv · 2 marks · Working and answer

Working and answer

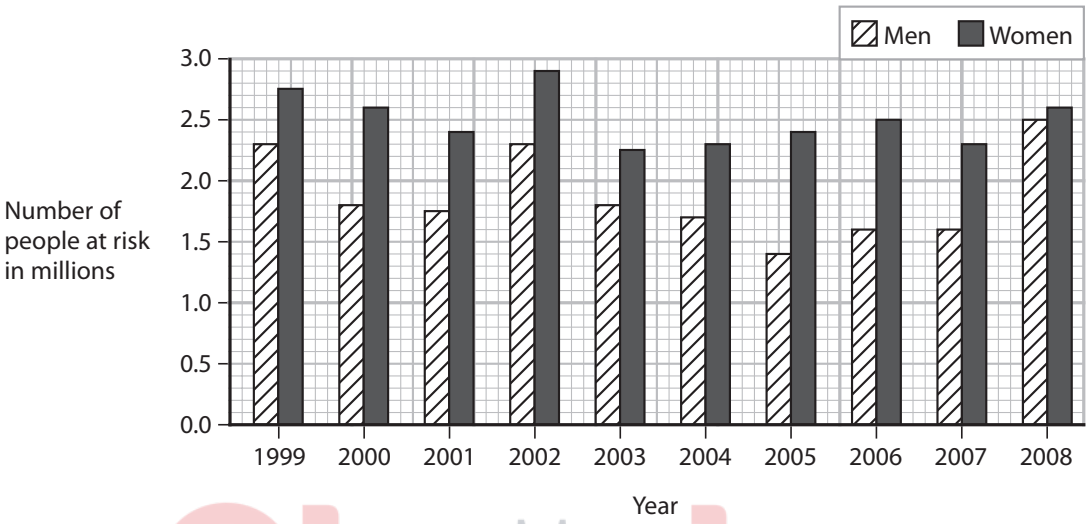
OLevels.com SCHOOL

6. Question 2 · part a.iv · 1 mark

June 2022 Paper 2

Further practice · Attempt part a.iv only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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P 7 0 9 7 4 A 0 4 2 0

6. Question 2 · part a.iv · 1 mark

June 2022 Paper 2

Further practice · Attempt part a.iv only.

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(iv) Suggest one possible reason why the number of females at risk of developing a mental illness increased from 2007 to 2008.

(1)

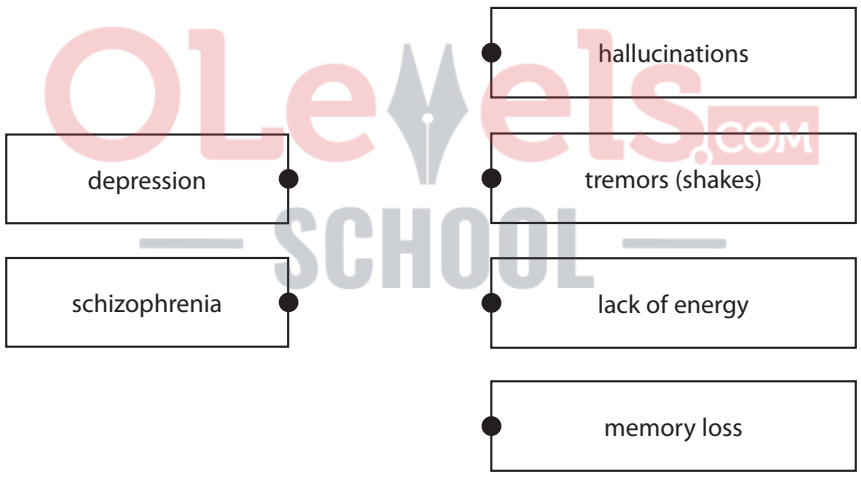
.....

.....

(b) The boxes list two mental illnesses and some symptoms.

Draw one straight line from each mental illness to a symptom of the illness.

(2)



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(c) Which combination of drugs is most likely to affect mental health?

(1)

- ☒ **A** alcohol and heroin
- ☒ **B** antibiotics and alcohol
- ☒ **C** cannabis and paracetamol
- ☒ **D** paracetamol and antibiotics

(d) People who use the drug cocaine often have a raised body temperature.

Name two body structures that help to control body temperature.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

6. Question 2 · part a.iv · 1 mark · Working and answer

Working and answer

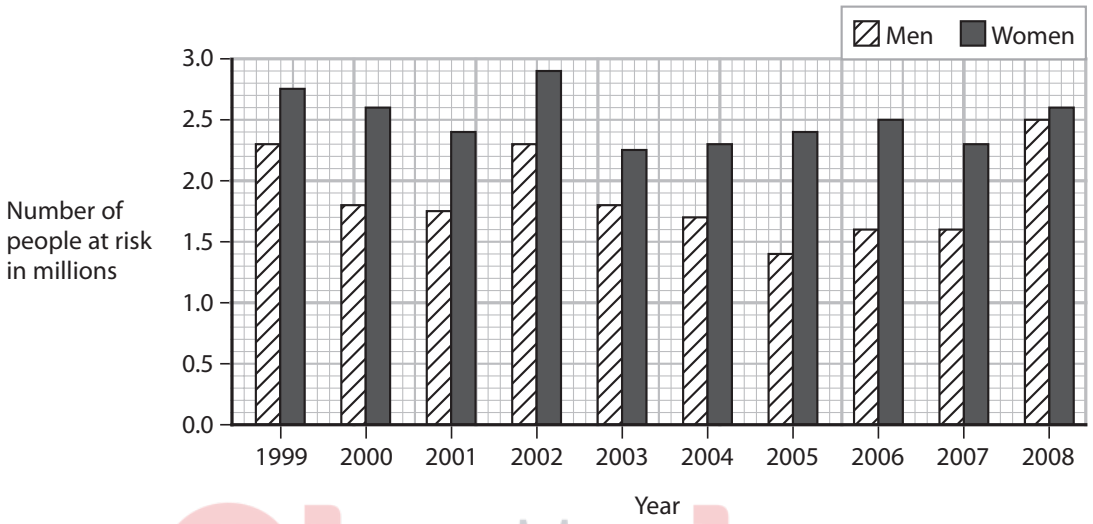
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7. Question 2 · part b · 2 marks

June 2022 Paper 2

Further practice · Attempt part b only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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P 7 0 9 7 4 A 0 4 2 0

7. Question 2 · part b · 2 marks

June 2022 Paper 2

Further practice · Attempt part b only.

(iv) Suggest one possible reason why the number of females at risk of developing a mental illness increased from 2007 to 2008.

(1)

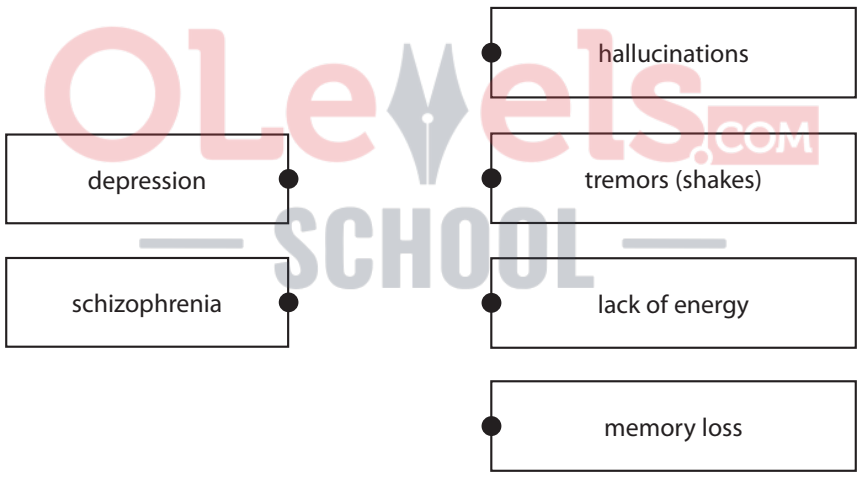
.....

.....

(b) The boxes list two mental illnesses and some symptoms.

Draw one straight line from each mental illness to a symptom of the illness.

(2)



(c) Which combination of drugs is most likely to affect mental health?

(1)

- ☒ **A** alcohol and heroin
- ☒ **B** antibiotics and alcohol
- ☒ **C** cannabis and paracetamol
- ☒ **D** paracetamol and antibiotics

(d) People who use the drug cocaine often have a raised body temperature.

Name two body structures that help to control body temperature.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

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7. Question 2 · part b · 2 marks · Working and answer

Working and answer

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Practice workbook · Identify ear parts and use the sound path to explain hearing and hearing loss

Naming ear structures, roles of pinna, eardrum, ossicles, cochlea, Eustachian tube and semicircular canals, and how glue ear, pressure, ageing or damage reduce hearing.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

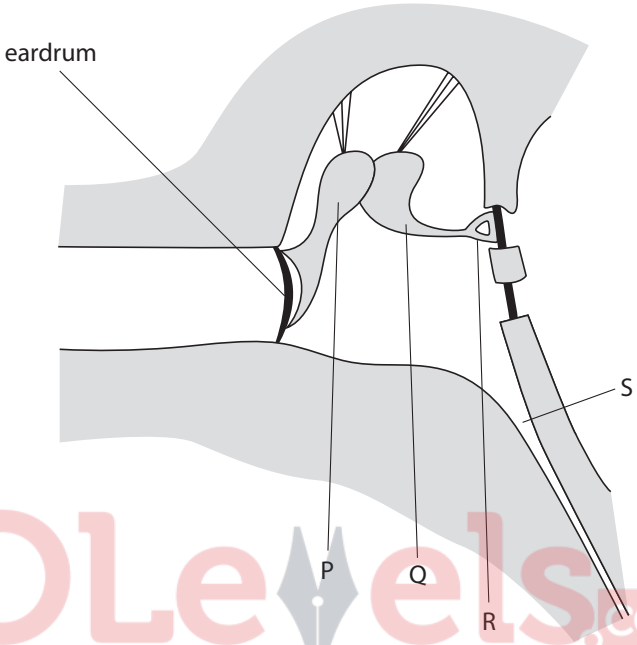
Type ID: HBI-EAR-PARTS-AND-HEARING-LOSS

1. Question 5 · part b.i · 4 marks

November 2020 Paper 2

Worked-example retry · Attempt part b.i only.

5 The diagram shows part of a human ear.



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(a) (i) Give the name of structure S.

(1)

.....

(ii) Name three parts of the ear that are **not** shown on the diagram.

(3)

1 .....

2 .....

3 .....

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1. Question 5 · part b.i · 4 marks

November 2020 Paper 2

Worked-example retry · Attempt part b.i only.

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(b) (i) Describe the functions of structures P, Q and R.

(4)

(ii) What is the name of the medium directly surrounding structures P, Q and R?

(1)

- ☒ **A** air
- ☒ **B** bone
- ☒ **C** fluid
- ☒ **D** oxygen

(c) Explain the importance of structure S in the process of hearing.

(3)

(Total for Question 5 = 12 marks)

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1. Question 5 · part b.i · 4 marks · Working and answer

Working and answer

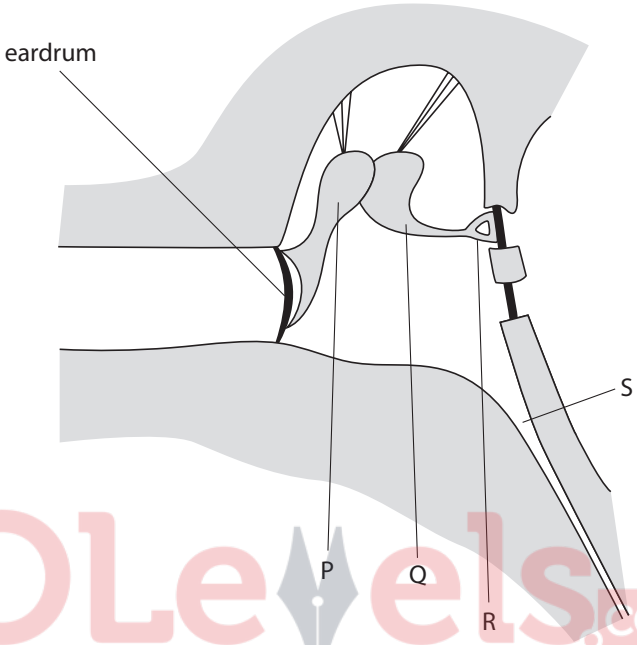
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2. Question 5 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

5 The diagram shows part of a human ear.



(a) (i) Give the name of structure S. **(1)**

.....

(ii) Name three parts of the ear that are **not** shown on the diagram. **(3)**

1 .....

2 .....

3 .....

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DO NOT WRITE IN THIS AREA

P 6 2 0 5 3 A 0 1 4 2 4

2. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

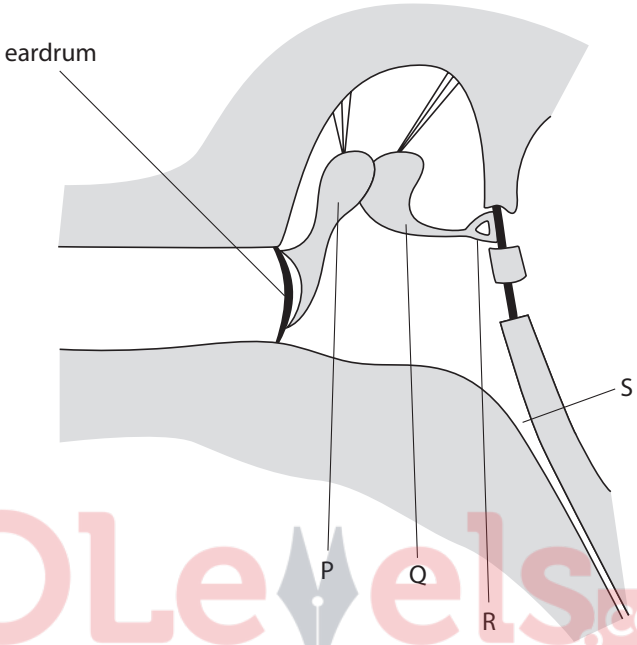
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3. Question 5 · part a.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

5 The diagram shows part of a human ear.



(a) (i) Give the name of structure S.

(1)

.....

(ii) Name three parts of the ear that are **not** shown on the diagram.

(3)

1 .....

2 .....

3 .....

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P 6 2 0 5 3 A 0 1 4 2 4

3. Question 5 · part a.ii · 3 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

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(b) (i) Describe the functions of structures P, Q and R.

(4)

(ii) What is the name of the medium directly surrounding structures P, Q and R?

(1)

- ☒ **A** air
- ☒ **B** bone
- ☒ **C** fluid
- ☒ **D** oxygen

(c) Explain the importance of structure S in the process of hearing.

(3)

(Total for Question 5 = 12 marks)

3. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

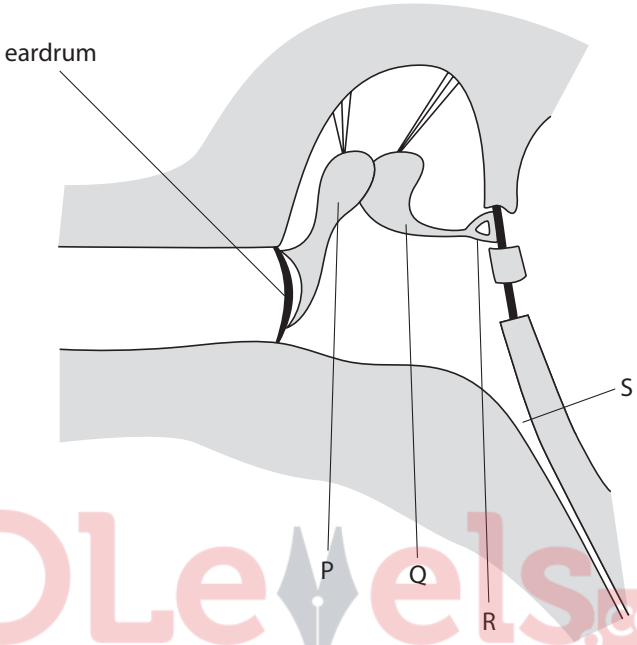
OLevels.com SCHOOL

4. Question 5 · part b.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part b.ii only.

5 The diagram shows part of a human ear.



(a) (i) Give the name of structure S.

(1)

.....

(ii) Name three parts of the ear that are **not** shown on the diagram.

(3)

1 .....

2 .....

3 .....

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P 6 2 0 5 3 A 0 1 4 2 4

4. Question 5 · part b.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(b) (i) Describe the functions of structures P, Q and R.

(4)

(ii) What is the name of the medium directly surrounding structures P, Q and R?

(1)

- ☒ **A** air
- ☒ **B** bone
- ☒ **C** fluid
- ☒ **D** oxygen

(c) Explain the importance of structure S in the process of hearing.

(3)

(Total for Question 5 = 12 marks)

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4. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

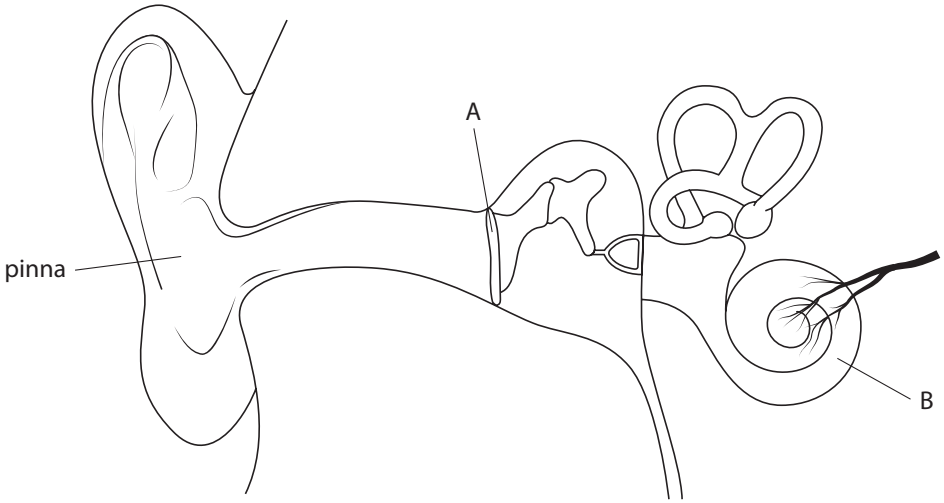
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5. Question 5 · part a.i · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.i only.

5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

.....

.....

.....

.....

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5. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

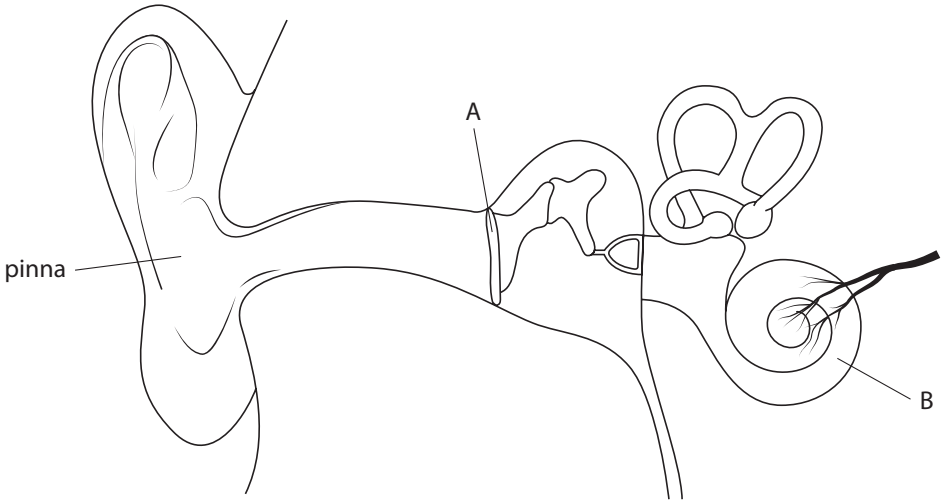
OLevels.com SCHOOL

6. Question 5 · part a.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.ii only.

5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

.....

.....

.....

.....

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OLevels.com SCHOOL

6. Question 5 · part a.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.ii only.

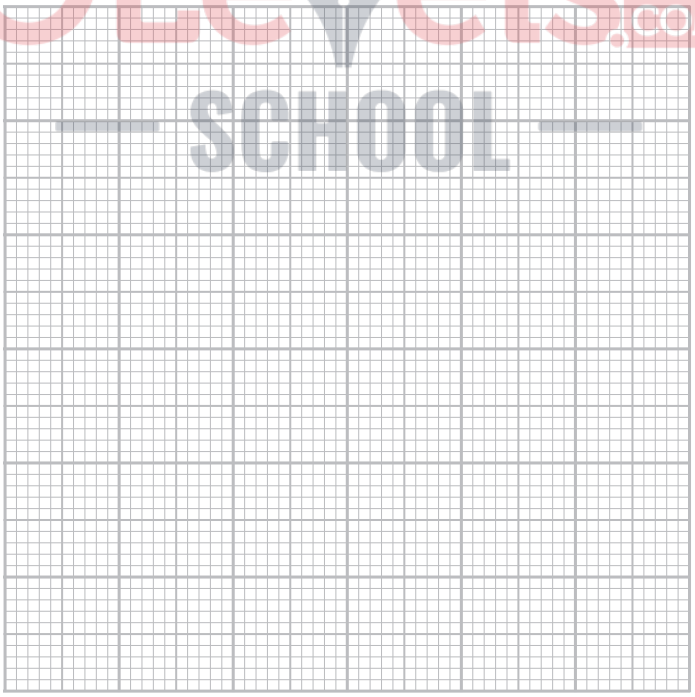
(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)



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6. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 5 · part b.iii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

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5 (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

(i) Describe an investigation a student could do to show if this is correct.

(3)

(ii) Give a reason why hearing loss occurs as people get older.

(1)

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7. Question 5 · part b.iii · 3 marks

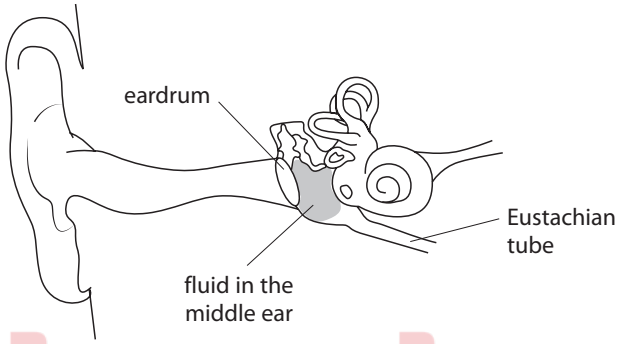
January 2023 Paper 1

Further practice · Attempt part b.iii only.

(b) Glue ear is a condition that can affect children and causes hearing loss.

Fluid accumulates in the middle ear, as shown in the diagram.

As the child grows, the slope of the Eustachian tube increases and the fluid drains.



(i) Explain how the fluid in the middle ear causes hearing loss.

(3)

(ii) Explain why children with glue ear get more bacterial ear infections than other children.

(2)

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P 7 2 2 7 1 A 0 1 4 2 4

7. Question 5 · part b.iii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

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(iii) Describe the role of the ear in maintaining a person's balance.

(3)

(Total for Question 5 = 12 marks)

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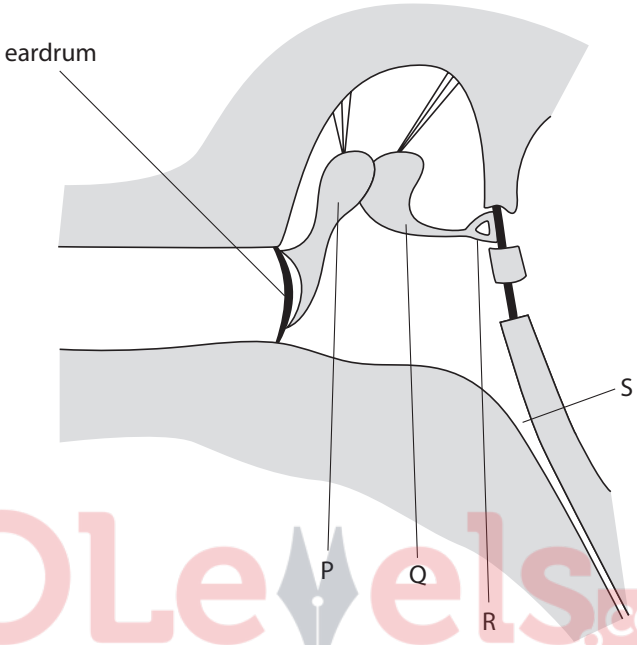
Working and answer

8. Question 5 · part c · 3 marks

November 2020 Paper 2

Further practice · Attempt part c only.

5 The diagram shows part of a human ear.



(a) (i) Give the name of structure S.

(1)

.....

(ii) Name three parts of the ear that are **not** shown on the diagram.

(3)

1 .....

2 .....

3 .....

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P 6 2 0 5 3 A 0 1 4 2 4

8. Question 5 · part c · 3 marks

November 2020 Paper 2

Further practice · Attempt part c only.

(b) (i) Describe the functions of structures P, Q and R.

(4)

(ii) What is the name of the medium directly surrounding structures P, Q and R?

(1)

- ☒ **A** air
- ☒ **B** bone
- ☒ **C** fluid
- ☒ **D** oxygen

(c) Explain the importance of structure S in the process of hearing.

(3)

(Total for Question 5 = 12 marks)

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8. Question 5 · part c · 3 marks · Working and answer

Working and answer

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9. Question 5 · part b.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

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5 (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

(i) Describe an investigation a student could do to show if this is correct.

(3)

(ii) Give a reason why hearing loss occurs as people get older.

(1)

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9. Question 5 · part b.i · 3 marks

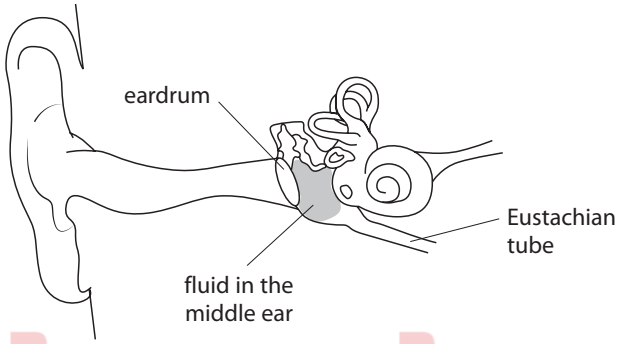
January 2023 Paper 1

Further practice · Attempt part b.i only.

(b) Glue ear is a condition that can affect children and causes hearing loss.

Fluid accumulates in the middle ear, as shown in the diagram.

As the child grows, the slope of the Eustachian tube increases and the fluid drains.



(i) Explain how the fluid in the middle ear causes hearing loss.

(3)

(ii) Explain why children with glue ear get more bacterial ear infections than other children.

(2)

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P 7 2 2 7 1 A 0 1 4 2 4

9. Question 5 · part b.i · 3 marks · Working and answer

Working and answer

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10. Question 5 · part a.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.ii only.

5 (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

(i) Describe an investigation a student could do to show if this is correct.

(3)

(ii) Give a reason why hearing loss occurs as people get older.

(1)

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10. Question 5 · part a.ii · 1 mark

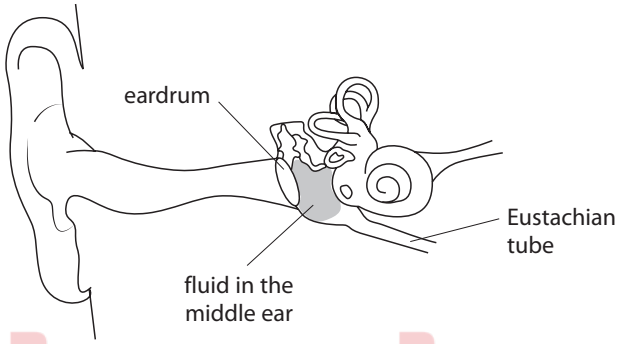
January 2023 Paper 1

Further practice · Attempt part a.ii only.

(b) Glue ear is a condition that can affect children and causes hearing loss.

Fluid accumulates in the middle ear, as shown in the diagram.

As the child grows, the slope of the Eustachian tube increases and the fluid drains.



(i) Explain how the fluid in the middle ear causes hearing loss.

(3)

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(2)

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10. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Show and explain why impulses travel one way across a synapse

Drawing the direction of an impulse across a synapse or through a reflex arc, marking the receptor end, and explaining or naming what keeps transmission one-way.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SHOW-EXPLAIN-ONE-WAY-IMPULSE

1. Question 6 · part b.iii · 4 marks

January 2022 Paper 2

Worked-example retry · Attempt part b.iii only.

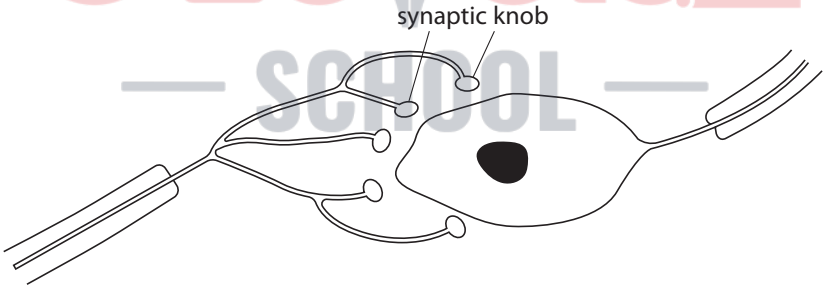
6 (a) Co-ordination of the activities of the body involves the nervous system and the hormonal system.

Complete the table to compare the two systems.

(3)

	Nervous system	Hormonal system
Route of transmission		
Type of signal		
Relative speed		

(b) The diagram shows a synapse.



(i) Draw an arrow on the diagram to show the direction an impulse travels.

(1)

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1. Question 6 · part b.iii · 4 marks

January 2022 Paper 2

Worked-example retry · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(ii) Explain how an impulse travels across a synapse.

(4)

DO NOT WRITE IN THIS AREA

(iii) Explain why an impulse can only travel in one direction across a synapse.

(4)

(Total for Question 6 = 12 marks)

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1. Question 6 · part b.iii · 4 marks · Working and answer

Working and answer

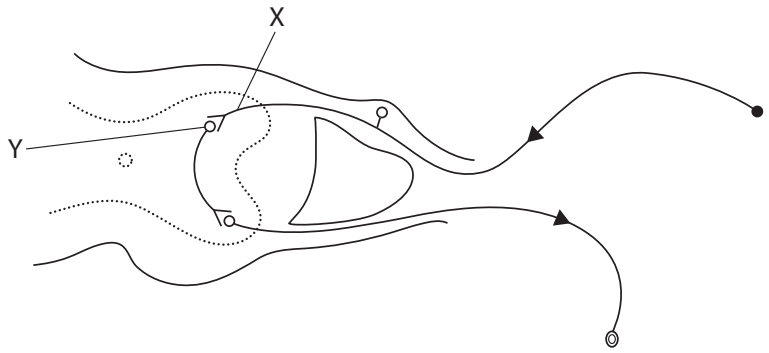
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2. Question 8 · part b · 1 mark

June 2021 Paper 1

Further practice · Attempt part b only.

8 The diagram shows a reflex arc involved in a reflex action.



(a) Describe the function of a reflex action.

(3)

(b) Using a line labelled Z, show on the diagram which part of the reflex arc would be connected to a receptor.

(1)

(c) Describe how the nerve impulse passes from position X to position Y in the reflex arc.

(6)

(Total for Question 8 = 10 marks)

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2. Question 8 · part b · 1 mark · Working and answer

Working and answer

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3. Question 3 · part b.ii · 1 mark

June 2025 Paper 2

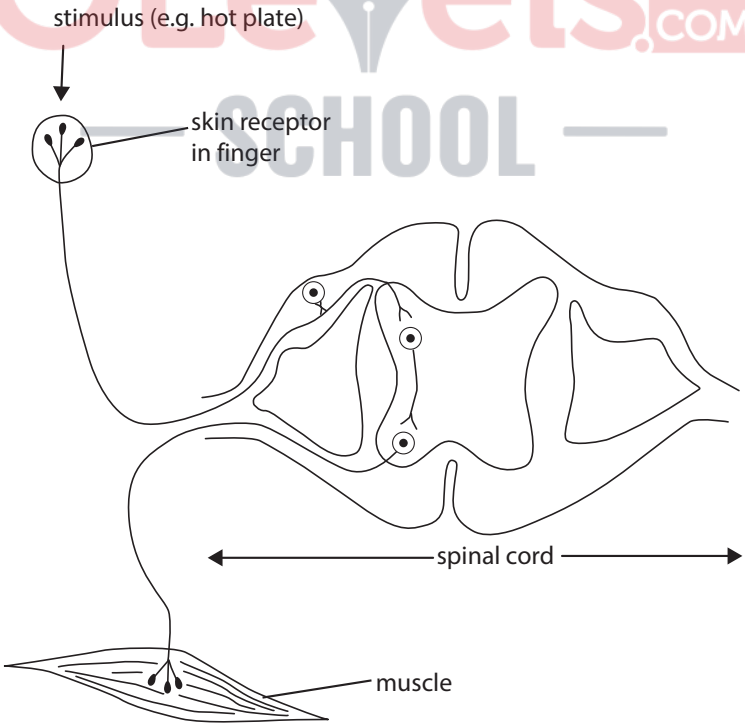
Further practice · Attempt part b.ii only.

3 (a) Describe the meaning of the term **reflex action**.

(3)

(b) The diagram shows a reflex arc.

The reflex arc contains three types of neurone.



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P 7 8 7 6 9 A 0 6 2 0

3. Question 3 · part b.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part b.ii only.

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(i) Use labels to identify each type of neurone in the diagram.

(3)

(ii) Draw an arrow on the diagram to show the direction of an impulse as it travels through the reflex arc.

(1)

(iii) State which structure in a reflex arc prevents the impulse travelling in the wrong direction.

(1)

.....

(Total for Question 3 = 8 marks)

Working and answer

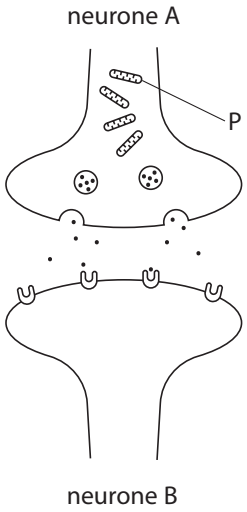
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4. Question 8 · part a.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part a.i only.

8 The diagram shows the gap between two neurones, A and B.



(a) (i) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

(ii) Explain the function of structure P in the transmission of the impulse across the gap. **(3)**

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P 6 2 0 5 4 A 0 2 2 2 4

4. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

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5. Question 6 · part b.i · 1 mark

January 2022 Paper 2

Further practice · Attempt part b.i only.

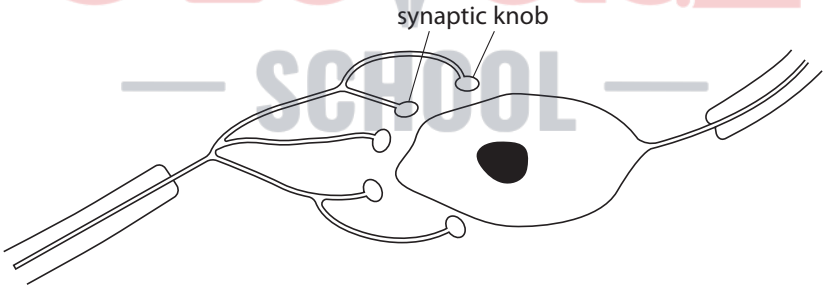
6 (a) Co-ordination of the activities of the body involves the nervous system and the hormonal system.

Complete the table to compare the two systems.

(3)

	Nervous system	Hormonal system
Route of transmission		
Type of signal		
Relative speed		

(b) The diagram shows a synapse.



(i) Draw an arrow on the diagram to show the direction an impulse travels.

(1)

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5. Question 6 · part b.i · 1 mark

January 2022 Paper 2

Further practice · Attempt part b.i only.

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(ii) Explain how an impulse travels across a synapse.

(4)

(iii) Explain why an impulse can only travel in one direction across a synapse.

(4)

(Total for Question 6 = 12 marks)

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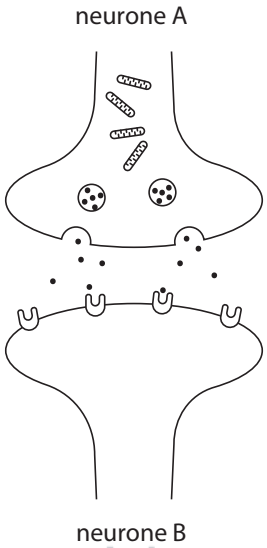
Working and answer

6. Question 3 · part a.ii · 1 mark

November 2024 Paper 2

Further practice · Attempt part a.ii only.

3 (a) The diagram shows the gap between two neurones, A and B.



(i) Name this gap. **(1)**

.....

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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P 7 5 9 5 9 A 0 8 2 4

6. Question 3 · part a.ii · 1 mark

November 2024 Paper 2

Further practice · Attempt part a.ii only.

(iii) Describe how a nerve impulse travels from one side of the gap to the other.

(6)

(iv) Explain why an impulse can only travel in one direction across the gap.

(3)

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6. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

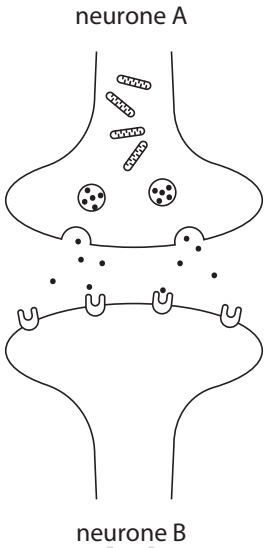
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7. Question 3 · part a.iv · 3 marks

November 2024 Paper 2

Further practice · Attempt part a.iv only.

3 (a) The diagram shows the gap between two neurones, A and B.



(i) Name this gap. **(1)**

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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P 7 5 9 5 9 A 0 8 2 4

7. Question 3 · part a.iv · 3 marks

November 2024 Paper 2

Further practice · Attempt part a.iv only.

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(iii) Describe how a nerve impulse travels from one side of the gap to the other.

(6)

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(iv) Explain why an impulse can only travel in one direction across the gap.

(3)

7. Question 3 · part a.iv · 3 marks

November 2024 Paper 2

Further practice · Attempt part a.iv only.

(b) Some people take the illegal drug known as cocaine.

State two dangers of taking cocaine.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 13 marks)

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Working and answer

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2

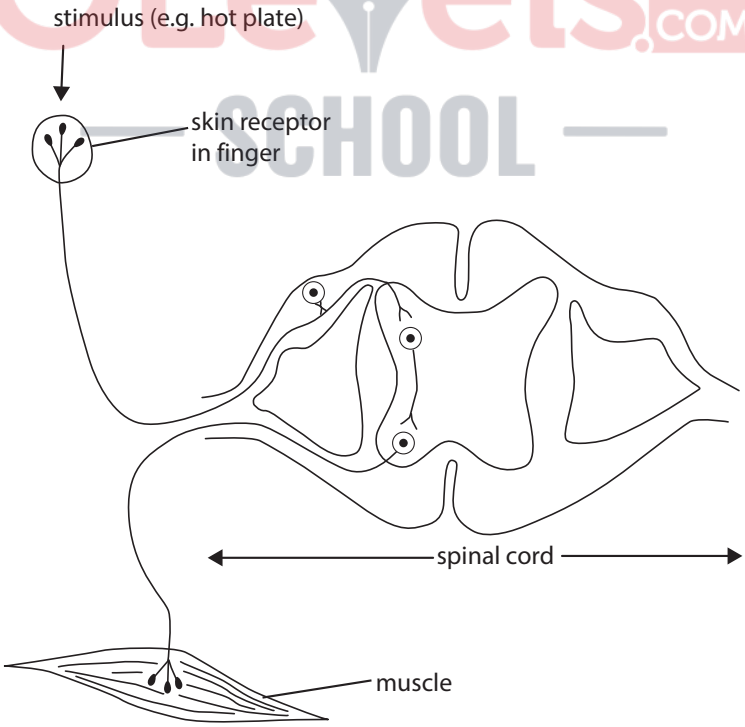
Further practice · Attempt part b.iii only.

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(3)

(b) The diagram shows a reflex arc.

The reflex arc contains three types of neurone.



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P 7 8 7 6 9 A 0 6 2 0

8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2

Further practice · Attempt part b.iii only.

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(i) Use labels to identify each type of neurone in the diagram.

(3)

(ii) Draw an arrow on the diagram to show the direction of an impulse as it travels through the reflex arc.

(1)

(iii) State which structure in a reflex arc prevents the impulse travelling in the wrong direction.

(1)

.....

(Total for Question 3 = 8 marks)

Working and answer

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